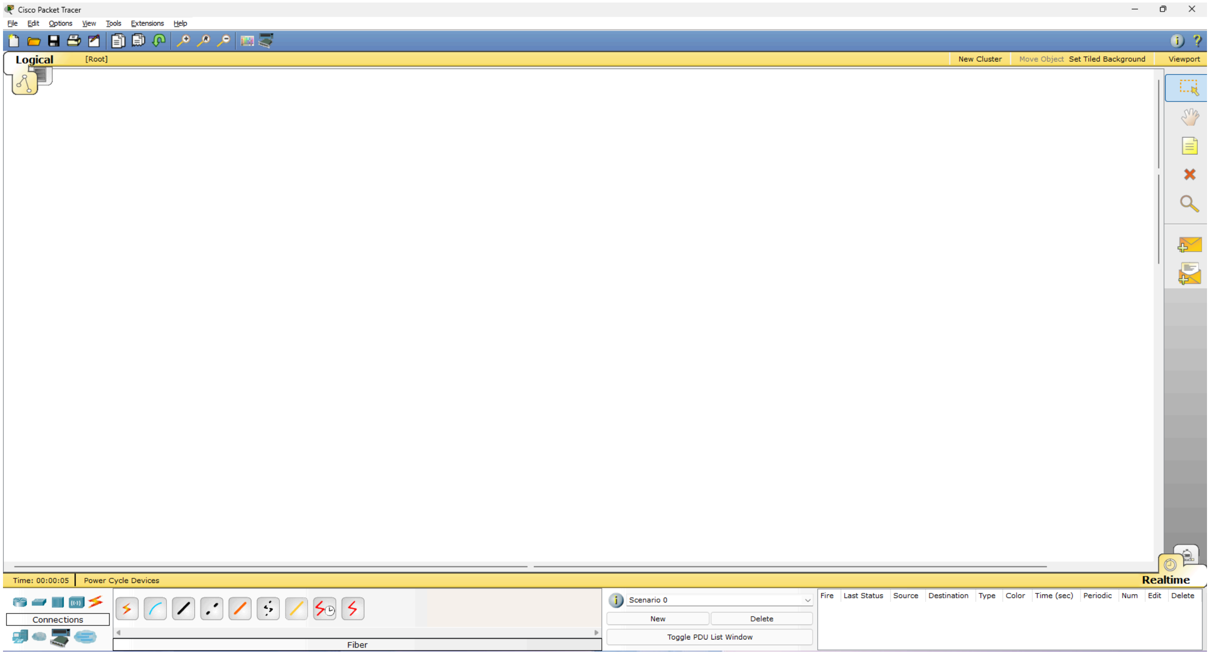
PRACTICALS

Computer Networks Laboratory

Practical 4.

Aim: To implement the LAN network using switch in packet tracer simulator

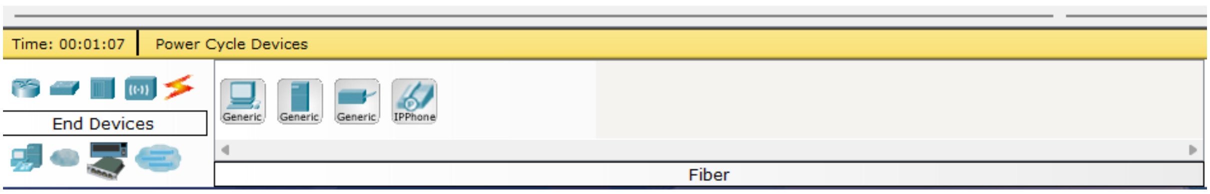
Step 1: Open Cisco packet tracer 5.1 as shown in **FIG 1**



[FIG 1]

<u>Step 2:</u> Select device to make LAN (Local Area Network) Network form the bottom Tool bar as shown in **FIG 2**
There are multiple devices used in this process are follows as:

1. Network Switch
2. Router
3. Hub
4. Wireless device
5. Cables/wires (for connection)
6. End devices (PC, Laptop, Server etc)
7. WAN Enulation
8. Custom made device
9. Multi User connection



[FIG 2]

<u>Step 3:</u> Select PCs, Network Switch, & connecting wires/Cables Shown in **FIG 3**

1. 3 - PCs
2. 1 - Network Switch
3. 3 - Connecting wires/ cables

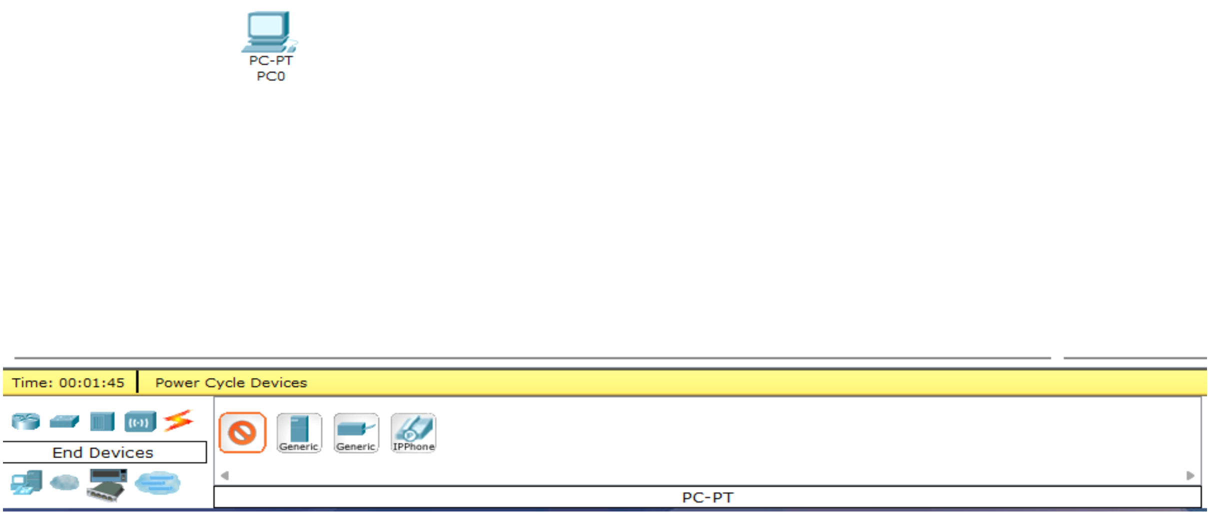


FIG 3

Step 4: Creating Network (Topology) Using selected devices as shown in **FIG 4**

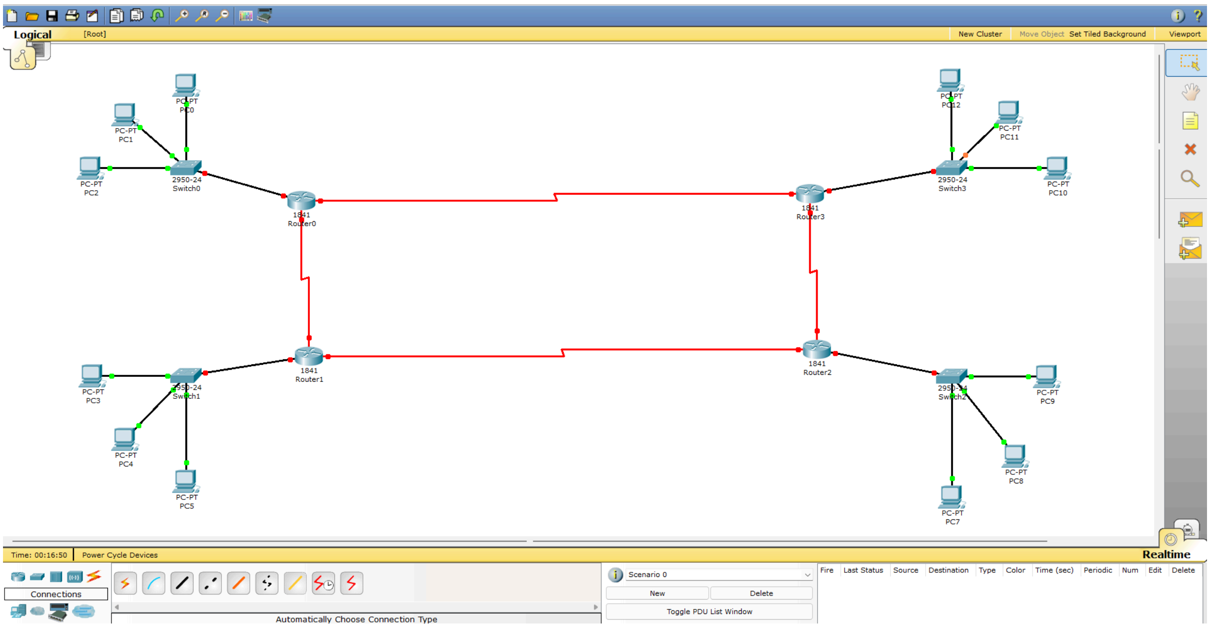


FIG 4

Step 5: Configuring The Setup with Network (static method) Using Default Gateway, Subnet Mask, Ip Address as shown in **FIG 5** and **FIG 6**

1. Click on PC
2. Go to Desktop Tab
3. Click on Ip Configuration

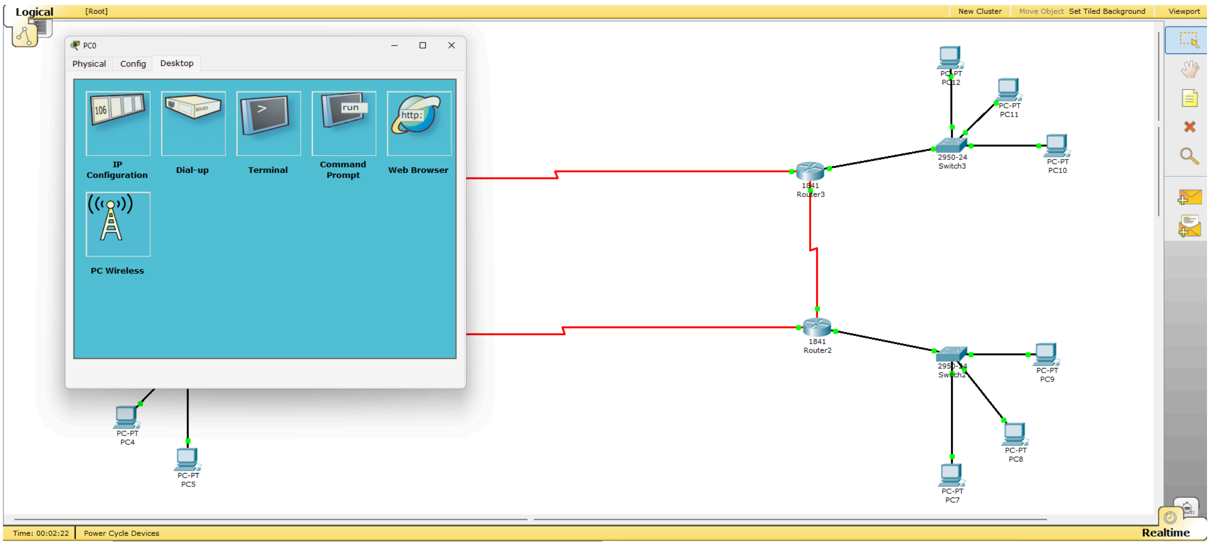


FIG 5

1. Select Or click on Static
2. Enter values as follows

3. Close the dialog box
4. Do the same process for all of the PCs

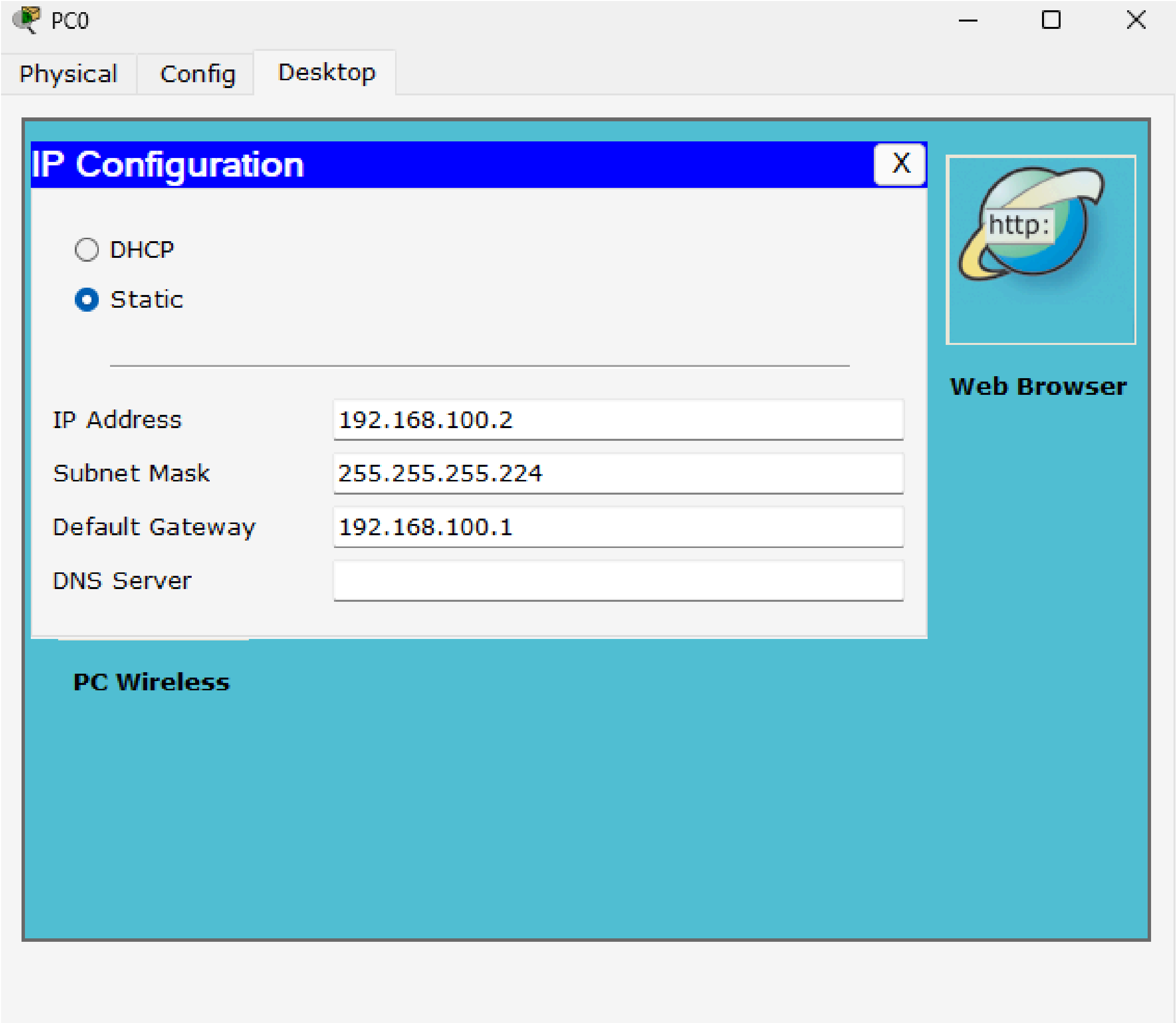


FIG 6

<u>Note:- IP for all pcs are as follows (with subnet mask)</u>

Common Subnet mask for all Pcs:- **255.255.255.224**

PC 0,1,2:- 192.168.100.2, 192.168.100.3, 192.168.100.4

PC 3,4,5:- 192.168.100.34, 192.168.100.35, 192.168.100.36

PC 6,7,8:- 192.168.100.66, 192.168.100.67, 192.168.100.68

PC 9,10,11:- 192.168.100.98, 192.168.100.99, 192.168.100.100

<u>Step 6:</u> Now Configuring Routers with following steps:

1. Click on Router
2. Click on Config tab
3. Click on FastEathernet0/0 for setup router and network switch
4. Click on serial0/0/0 &serial0/0/0 for setup routers

<u>Step 7:</u> setup the values given in below **FIG 7** for FastEathernet 0/0

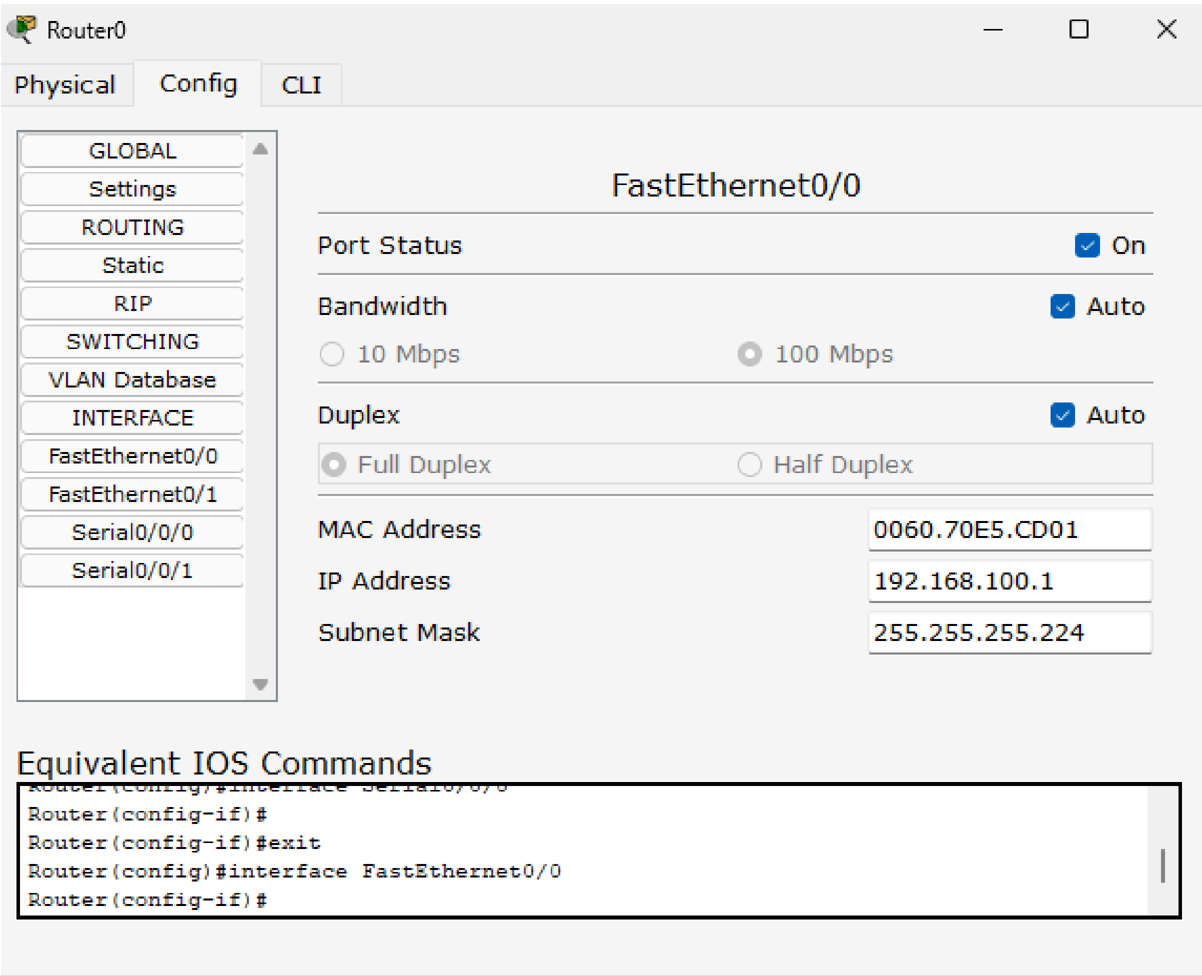


FIG 7

<u>Step 8:</u> setup the values given in below **FIG 8** for serial0/0/0 & **FIG 9** for serial0/0/1

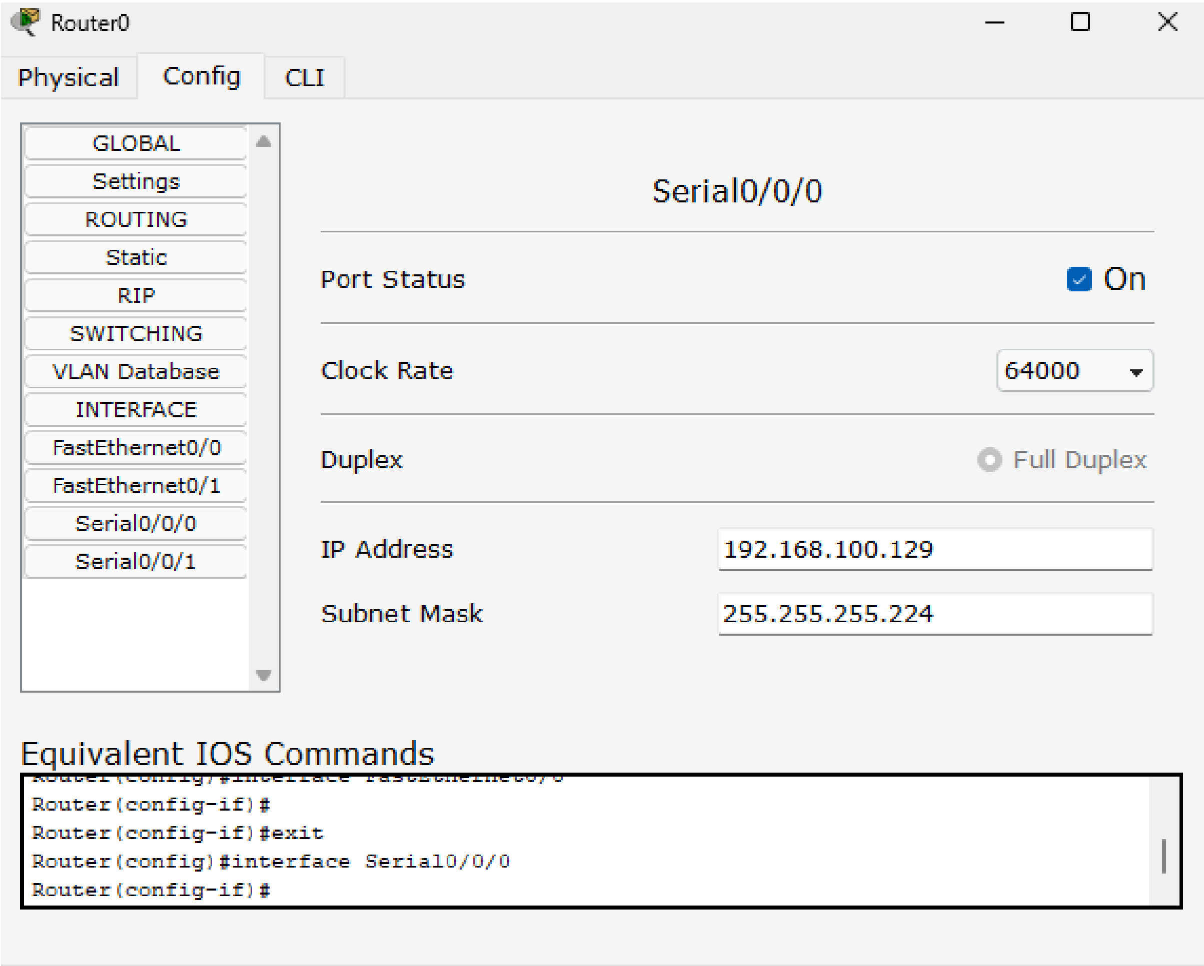


FIG 8

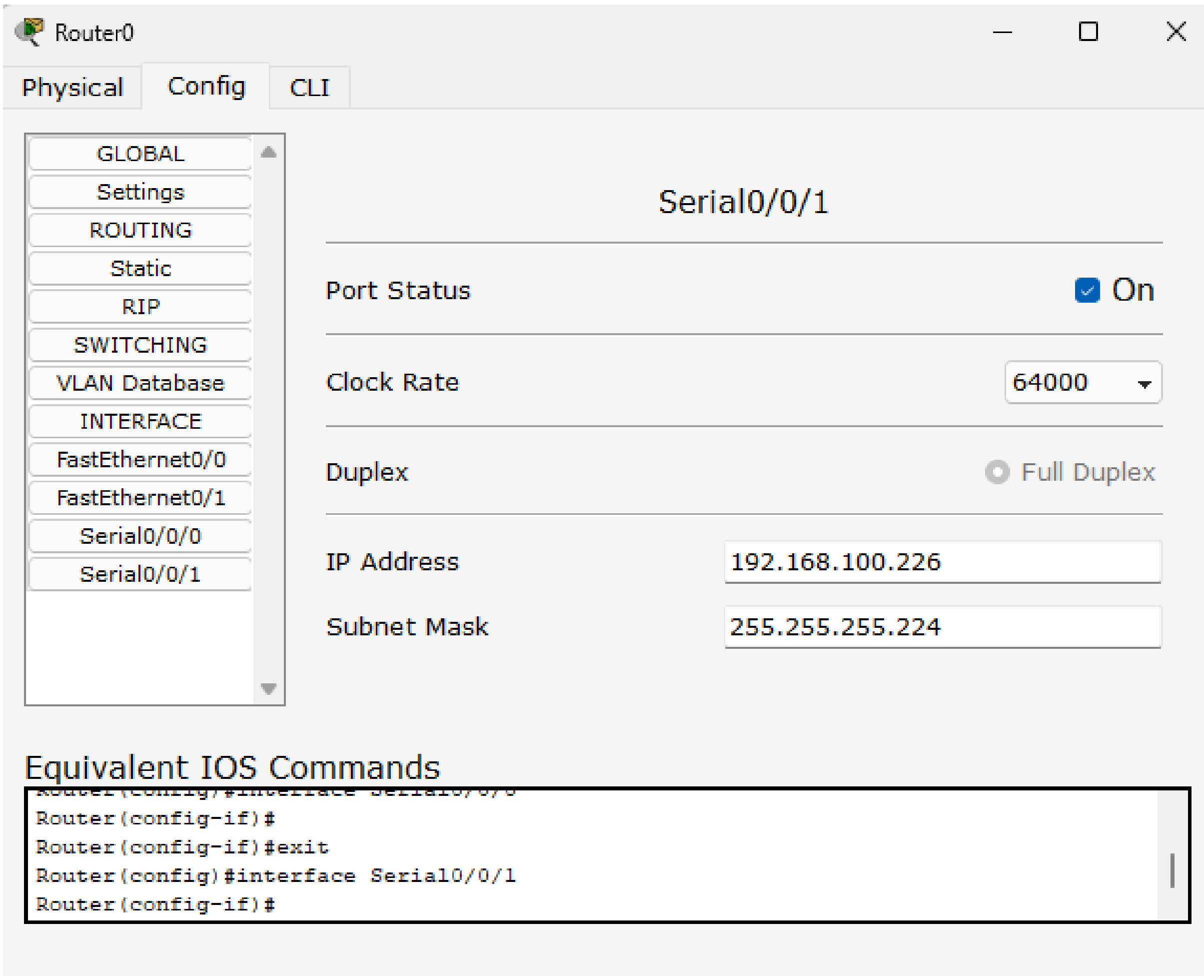


FIG 9

For other router refer the following table

Router 0

	FastEathernet0/0	Serial0/0/0	Serial0/0/1
Port Status	ON	ON	ON
Clock Rate	–	64000	64000
IP ADDRESS	192.168.100.1	192.168.100.129	192.168.100.226
SUBNET MASK	255.255.255.224	255.255.255.224	255.255.255.224

Router 1

	FastEathernet0/0	Serial0/0/0	Serial0/0/1
Port Status	ON	ON	ON
Clock Rate	–	64000	64000
IP ADDRESS	192.168.100.33	192.168.100.130	192.168.100.161
SUBNET MASK	255.255.255.224	255.255.255.224	255.255.255.224

Router 2

	FastEathernet0/0	Serial0/0/0	Serial0/0/1
Port Status	ON	ON	ON
Clock Rate	–	64000	64000
IP ADDRESS	192.168.100.65	192.168.100.193	192.168.100.162
SUBNET MASK	255.255.255.224	255.255.255.224	255.255.255.224

Router 3

	FastEathernet0/0	Serial0/0/0	Serial0/0/1
Port Status	ON	ON	ON
Clock Rate	–	64000	64000
IP ADDRESS	192.168.100.97	192.168.100.194	192.168.100.225
SUBNET MASK	255.255.255.224	255.255.255.224	255.255.255.224

<u>Step 9:</u> Setup RIP Static Routes follow the following steps

1. Open Router
2. Go to config tab
3. Under STATIC you will find Static click on it and fill the values
4. For router refer the following tables

Router 0

Destination	Mask	Next Hop
192.168.100.32	255.255.255.224	192.168.100.130
192.168.100.64	255.255.255.224	192.168.100.130
192.168.100.96	255.255.255.224	192.168.100.225
192.168.100.160	255.255.255.224	192.168.100.130
192.168.100.192	255.255.255.224	192.168.100.225

Router 1

Destination	Mask	Next Hop
192.168.100.0	255.255.255.224	192.168.100.129
192.168.100.64	255.255.255.224	192.168.100.162
192.168.100.96	255.255.255.224	192.168.100.129
192.168.100.192	255.255.255.224	192.168.100.162
192.168.100.224	255.255.255.224	192.168.100.129

Router 2

Destination	Mask	Next Hop
192.168.100.0	255.255.255.224	192.168.100.161
192.168.100.32	255.255.255.224	192.168.100.161
192.168.100.96	255.255.255.224	192.168.100.194
192.168.100.128	255.255.255.224	192.168.100.161
192.168.100.224	255.255.255.224	192.168.100.194

Router 3

Destination	Mask	Next Hop
192.168.100.0	255.255.255.224	192.168.100.226
192.168.100.32	255.255.255.224	192.168.100.226
192.168.100.64	255.255.255.224	192.168.100.193
192.168.100.128	255.255.255.224	192.168.100.226
192.168.100.160	255.255.255.224	192.168.100.193

<u>Step 10:</u> Try to send Packets Through-out the network

At first attempt it will failed (to share outside the network) but in second attempt of sending packets it will success as shown in **FIG8**

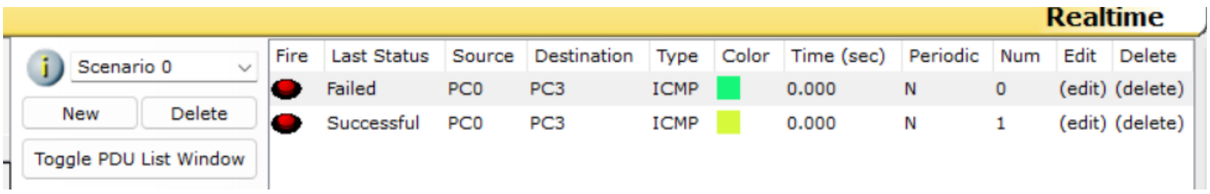


FIG 8